

Attention Is All You Need

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- **Background:** Dominant sequence transduction models (e.g., RNNs, LSTMs) rely on complex recurrent or convolutional neural networks.
- **Limitations:**
 - **RNNs:** Sequential nature precludes parallelization within training examples, hindering scalability for longer sequences.
 - **CNNs:** Computational cost to relate signals from two distant positions grows with distance (linearly or logarithmically).
- **Core Contribution:** Propose the **Transformer**, the first transduction model relying entirely on **Self-Attention** to compute input/output representations without recurrence or convolutions.

Theoretical Framework: Scaled Dot-Product Attention

Attention Mechanism

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

- **Q, K, V**: Query, Key, and Value vectors.
- **Scaling Factor** $\frac{1}{\sqrt{d_k}}$: Prevents the dot product from growing too large.
- **Efficiency**: Optimized matrix multiplication.

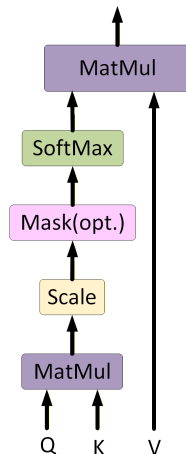


Fig 1: Scaled Dot-Product Attention

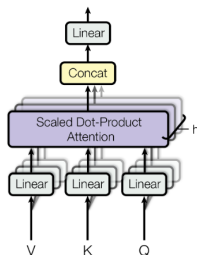
Theoretical Framework: Multi-Head Attention

- **Concept:** Allows the model to jointly attend to information from different representation subspaces at different positions.
- **Mechanism:**

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O$$

where $\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$.

- **Advantage:** Enables the model to capture complex semantic and syntactic dependencies in parallel.



Model Architecture

- **Encoder-Decoder:**

Stacked $N = 6$ identical layers for both modules.

- **Residual Connections:**

Applied to each sub-layer followed by Layer

Normalization:

$LayerNorm(x + Sublayer(x))$.

- **Positional Encoding:**

Uses Sine/Cosine functions to inject information about the relative or absolute position of the tokens.

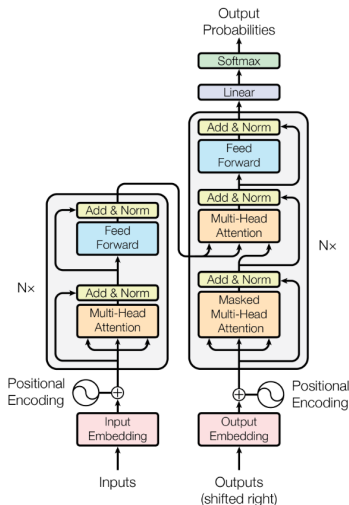


Fig 2: The Transformer Model Architecture

Results: Performance & Efficiency

- **Translation Quality:** Achieves **28.4 BLEU** on WMT 2014 En-De, establishing a new state-of-the-art result.
- **Training Cost:** Significantly more parallelizable; reaches SOTA with a fraction of the training time compared to previous models.

Table: Translation Performance and Training Cost

Model	EN-DE (BLEU)	EN-FR (BLEU)	Training Cost (FLOPs)
ByteNet	23.75	-	1.0×10^{20}
GNMT + RL	24.6	39.92	2.3×10^{19}
Transformer (base)	27.3	38.1	3.3×10^{18}
Transformer (big)	28.4	41.8	2.3×10^{19}

Discussion: Self-Attention Visualizations

- **Anaphora Resolution:**

Different attention heads learn to resolve pronouns (e.g., "its") by attending to the correct nouns.

- **Syntactic Structure:** Model captures long-range dependencies, such as the relationship between a verb and its distant object.

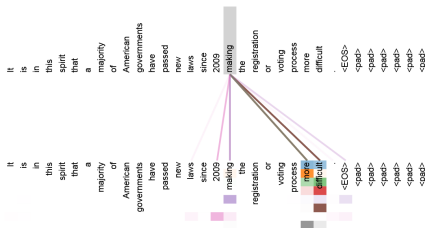


Fig 3: Attention behavior for anaphora resolution

- **Summary:**

- ① The Transformer is the first model to rely entirely on self-attention for sequence transduction.
- ② It achieves superior translation quality while being faster to train and more parallelizable.

- **Future Work:**

- Applying Transformer to other modalities (e.g., images, audio).
- Investigating local, restricted attention mechanisms for very large inputs.

Thank you for your attention!